

PRIZES



¥

20,000



¥

10,000



¥

8,000

Participant Guidebook

AI + Materials Competition

Intelligent Innovation of Materials, Empowering the Future

Token and Computing Power Platform Portal:

Volcengine: https://www.volcengine.com/activity/codingplan?utm_campaign=20260520&utm_content=codingplan&utm_medium=school&utm_source=qinghua&utm_term=codingplan

Alibaba: <https://university.aliyun.com/action/ai-materials>

Paratera: <https://ai.paratera.com>

Sugon: https://www.scnet.cn/ui/acm/index.html#/activity?invitation_code=rz0NP2gq

Ctyun: <https://ctxirang.ctyun.cn/maas/home>

Organizers

School of Materials Science and Engineering, Tsinghua University

School of Materials Science and Engineering, Peking University

School of Materials Science and Engineering, Beihang University

School of Materials Science and Engineering, Beijing Institute of Technology

School of Materials Science and Engineering, University of Science and Technology Beijing

2026 Beijing, China

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I. Competition Objectives

The 2026 AI+ Materials Competition focuses on the cutting-edge trends in new material R&D and industrial transformation. We expect this competition could provide an interdisciplinary innovation platform aiming the following objectives:

- ✧ Foster innovative solutions and talents for AI-empowered materials science.
- ✧ Promote a deep integration of artificial intelligence technology with the materials science and engineering.
- ✧ Bridge the gap and accelerate the transition from university research to industrial applications.
- ✧ Create an AI+ practice ecosystem for materials-related students and employees.

II. Participants and Team Requirements

Participants: Full-time post-secondary, secondary school students and employees from industry are welcome.

Team Requirements: Both individual and team participations are acceptable. No one is allowed to serve as the team leader for more than one team.

III. Tracks and Competition Content

The competition features six tracks. Each team may submit work to only one track.

Track 1: AI + Materials Design and R&D

This track focuses on leveraging artificial intelligence to accelerate the discovery, design, and performance optimization of new materials, thereby shortening R&D cycles and reducing trial-and-error costs.

Specific Directions:

- ✧ Inverse Materials Design: Utilizing generative AI (e.g., Generative Adversarial Networks, Diffusion Models), Graph Neural Networks (GNNs), etc., to inversely generate candidate material molecules or crystal structures based on target properties (e.g., strength, thermal conductivity, electrochemical performance).
- ✧ Materials Science Knowledge Graphs and Patent Analysis: Construction of knowledge graphs that integrate material composition, structure, properties, processes, and patent literature to enable intelligent retrieval, technology roadmap mining, and innovation trend analysis.
- ✧ High-Throughput Computing and Machine Learning Force Fields: Development of machine learning force field models to replace traditional first-principles calculations, enabling large-scale, high-efficiency simulation and screening of material microscopic behaviors.
- ✧ Intelligent Analysis of Materials Characterization Data: Applying deep learning algorithms to characterization data (such as XRD, SEM, and TEM) for automatic pattern recognition, phase analysis, defect detection, and

results extraction.

Submission Requirements:

Competition Application Form (to be completed using the template provided by the Organizing Committee), Solution Report, Algorithm/Code, Demonstration Program (Demo), Validation Dataset, Experimental Samples, etc.

Track 2: AI + Teaching

This track aims to promote the deep integration of AI technology with the development of materials science education, innovating teaching models and enhancing educational effectiveness.

Specific Directions:

- ✧ AI-Enhanced Teaching Innovation Application Cases: Demonstrating how AI tools can be conjugated with learning analytics, personalized resource recommendation, virtual simulation experiments, and intelligent tutoring to optimize instructional design, implementation, and evaluation processes for materials science courses.
- ✧ AI-Integrated Digital Course Development: Developing online courses or teaching modules for materials science that integrate AI interactive content and intelligent assessment systems, exploring innovative teaching models

such as exploratory learning and project-based learning..

- ✧ Digital Literacy Enhancement and Micro-lesson Examples: Focusing on a single knowledge point, using AI tools to create micro-lessons and submit a complete package including "short form content + Teaching Improvement Plan + Reflection on AI Tool Application" to support the transition of AI from competition projects to regular classroom application.
- ✧ Intelligent Agent Development: Focusing on the development of intelligent agents in an education setting, constructing intelligent platforms covering modules such as teaching data management, personalized learning services, and curriculum collaboration to achieve integrated teaching support.

Submission Requirements:

Competition Application Form (to be completed using the template provided by the Organizing Committee), Instructional Design Plan, Teaching Reflection Report, Lecture Video (within 10 minutes), Classroom Recording (within 10 minutes), Courseware, Course Resource Package, Intelligent Agent, etc.

Track 3: AI+Smart Manufacturing of Materials

This track focuses artificial intelligence applications in material production and manufacturing processes to achieve smart factories, flexible production, and sustainable development.

Specific Directions:

- ✧ Intelligent Production Scheduling and Process Optimization: Utilizing AI algorithms for modeling, real-time optimization, and adaptive control of material preparation processes (e.g., smelting, rolling, heat treatment) to enhance production efficiency and consistency.
- ✧ Intelligent Equipment Maintenance and Predictive Maintenance: Employing machine learning models based on IoT data for equipment monitoring, fault prediction, and diagnosis to reduce the risk of unplanned downtime.
- ✧ Supply Chain Collaboration and Intelligent Scheduling: Applying AI for market demand forecasting, raw material procurement optimization, inventory management, and logistics scheduling to achieve agile supply chain response.

Submission Requirements:

Competition Application Form (to be completed using the template provided by the Organizing Committee), Solution PPT, Technical Validation Video (within 10 minutes), Simulation Model, Data Analysis Report, etc.

Track 4: AI+Materials Testing and Characterization

This track concentrates on leveraging AI to develop integrated intelligence

in material quality inspection, performance evaluation, and failure analysis.

Specific Directions:

- ✧ Industrial Vision and Non-Destructive Testing: Developing systems based on machine vision and deep learning for automatic online detection and classification of material surface defects (e.g., cracks, scratches, inclusions).
- ✧ Intelligent Quality Analysis and Traceability: Integrating entire production process data and using AI to establish correlation models between product quality and process parameters, enabling root cause analysis and full lifecycle traceability of quality issues.
- ✧ Multi-Scale Characterization Data Fusion Analysis: Fusing multi-source, multiscale characterization data (e.g., spectroscopy, imaging, mechanical properties) through multimodal AI models to achieve comprehensive assessment and prediction of material performance and structure.

Submission Requirements:

Competition Application Form (to be completed using the template provided by the Organizing Committee), Inspection System Proposal, Algorithm/Model, Test Dataset, Analysis Report, etc.

Track 5: AI+Corporate Innovation and Applications

This track directly addresses industrial needs. It may adopt an "Open

Bidding" format where enterprises release specific challenges, or it may feature open innovation themes.

Submission Requirements:

Competition Application Form (to be completed using the template provided by the Organizing Committee). Additional materials should be designed by the participating teams based on the specific enterprise challenge.

Track 6: AI + Integrated Applications

This track is an open category encouraging innovative thinking across disciplines, professions, and fields. There are no restrictions on the specific professional background of the submissions; however, the work must demonstrate clear innovation, application potential, and a certain level of technical depth.

Specific Directions:

Including but not limited to:

- ✧ Interdisciplinary Exploration: Innovative applications or research proposals resulting from the deep integration of AI with multiple disciplines (biology, chemistry, environmental science, energy, architecture, electronics and information, economics, humanities, arts, etc.).
- ✧ Social and Public Services: Innovative solutions applying AI to public

service domains such as smart cities, healthcare, educational technology, financial technology, legal technology, etc.

- ✧ Infrastructure Tool Development: Developing innovative algorithms, frameworks, datasets, development tools, productivity software, or PPT creation software for AI developers or researchers, aimed at enhancing the overall efficiency of AI R&D and application

Submission Requirements:

Competition Application Form (to be completed using the template provided by the Organizing Committee), Algorithm/Code, Demonstration Program (Demo), Test Data, Application Cases, or other materials demonstrating the work's innovation and value.

IV. Competition Schedule

April 2026: Launch and Registration

The Organizing Committee will announce the competition and detailed challenge descriptions, organizing online briefings and technical training. Participating teams must complete registration by May 31th (late submissions will not be accepted).

May-July 2026: Project Development and Submission

Participating teams will develop their projects. The Organizing Committee will organize technical discussion forums and expert mentoring sessions.

Teams must submit their final work by July 31st (late submissions will be considered forfeiture).

Early August 2026: Preliminary Review

The competition committee will collect and evaluate the applications, and select the finalists.

Mid-August 2026: Tech Innovation Bootcamp

A two-day, offline, closed-door training camp focused on enhancing pitching skills and business commercialization capabilities will be provided. Participants who successfully complete the training will receive a certificate and priority to access the final competition's micro-pitch session.

Late August 2026: Final Competition

An on-site demonstration with a 6-minute presentation + 2 minute Q&A will be evaluated. The panel of judges includes experts in academic, industry and venture capital.

September 2026: Award Ceremony

An on-site award ceremony will be celebrated (The date and venue will be announced later). All participating teams are welcome to attend.

V. Organizing Committee

Chair: Hao Chen

Deputy Chair: Xiumei Wang

Members: Huijun Chen, Ping Du, Xuning Feng, Zhengcao Li, Jianbo Liu, Xingtao Liu, Ruitao Lyu, Hui Qiao, Yutian Shen, Cheng Song, Weipeng Wang, Xiaonan Wang, Yongzheng Wen, Xiaoqing Xi, Mingyong Xu, Di Yi, Ming Zhao

Secretary-General: Yang Shao

Deputy Secretaries-General: Tao Lin, Jiali Zhao

VI. Judging Committee

Chair: Yuanhua Lin

Deputy Chair: Hao Chen

Members: Xiang Chen, Hongbiao Dong, Yanhao Dong, Peixun Fan, Qianming Gong, Shanfang Huang, Jinwu Kang, Cheng Song, Xiaonan Wang, Yongzheng Wen, Jiayi Yan

VII. PRIZES FOR EACH CATEGORY

RMB 20,000 - First-class award

RMB 10,000 - Second-class award

RMB 8,000 - Third-class award

VIII. Scoring Criteria

Track 1: AI + Materials Design and R&D

Dimension	Weight	Specific Content	Scoring Criteria
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Technical Innovation	30%	Originality of model, advancement of methodology, suitability for materials problem	Significant innovation (25–30) Clear improvement/innovation (18–24) Adaptation of existing methods with adjustments (10–17) Conventional approach, insufficient innovation (0–9)
Scientific Rigor	20%	Theoretical basis, data quality, experimental design rationality, credibility of validation methods	Robust theory, complete data, thorough validation (16–20) Basic scientific basis, generally credible data (11–15) Insufficient basis, weak validation (6–10) Obvious scientific issues (0–5)
Completeness & Feasibility	20%	Code/model operability, Demo effectiveness, report/documentation completeness	Complete system, vivid demonstration, comprehensive materials (16–20) Basic functionality, relatively clear demonstration (11–15) Missing functions, simple demonstration (6–10) Cannot be demonstrated or incomplete materials (0–5)
Application Value & Potential	15%	Ability to solve practical materials problems, technological transferability, impact on research paradigms	Significant value, generalizable, paradigm impact (12–15) Some application potential (8–11) Limited application scope (4–7) Unclear value (0–3)
Presentation & Team Collaboration	15%	Persuasiveness of presentation, demonstration of team division, collaboration in Q&A	Engaging presentation, cohesive team collaboration (12–15) Clear expression, good collaboration (8–11) Average expression, fair collaboration (4–7) Unclear expression, insufficient collaboration (0–3)

Track 2: AI+Education

Dimension	Weight	Specific Content	Scoring Criteria
Educational Philosophy & Objectives	30%	Educational orientation of instructional design, clarity of objectives, integration of educational concepts with AI	Advanced philosophy, clear objectives, natural integration (24–30) Relatively clear objectives, reasonable integration (18–24) Vague objectives, forced integration (10–17) Outdated or missing philosophy (0–9)
Depth of AI Integration & Teaching Innovation	25%	Appropriateness of AI tool usage, restructuring of teaching process, level of interaction and personalization	Deep integration, novel model, strong interactivity (19–25) Good integration, some innovation (13–18) Simple application, limited innovation (7–12) Superficial or inappropriate integration (0–6)
Teaching Effectiveness & Scalability	20%	Student feedback/data effects, replicability, practical contribution to teaching improvement	Significant effect, easily scalable, evidence-based (16–20) Good effect, some scalability (11–15) Average effect, limited scalability (6–10) Unclear effect or difficult to implement (0–5)
Depth of Reflection & Continuous Improvement	10%	Critical thinking on AI tool application, rationality of teaching improvement plans	Profound reflection, systematic and feasible improvement plan (9–10) Some reflection and improvement ideas (7–8) Superficial reflection, vague direction (4–6) Lack of reflection (0–3)
Presentation & Team Collaboration	15%	Persuasiveness of presentation, demonstration of team division, collaboration in Q&A	Engaging presentation, cohesive team collaboration (12–15) Clear expression, good collaboration (8–11) Average expression, fair collaboration (4–7) Unclear expression, insufficient collaboration (0–3)

Track 3: AI+Smart Manufacturing of Materials

Dimension	Weight	Specific Content	Scoring Criteria
Solution Feasibility	30%	Clear technical roadmap, reliable simulation/validation data, controllable implementation cost	Fully feasible, thorough validation, clear cost advantage (25–30) Basically feasible, some validation support (18–24) Partially feasible, insufficient validation (10–17) Feasibility questionable (0–9)
Process Optimization & Efficiency Improvement	25%	Actual impact on production process, improvement in capacity/quality/energy consumption, etc.	Significant improvement, clear quantifiable indicators (20–25) Some improvement, relatively clear indicators (13–19) Limited improvement, vague indicators (6–12) No significant optimization (0–5)
System Completeness & Integration	15%	Complete system architecture, good module synergy, compatibility with production environment	Complete system, high integration, strong compatibility (12–15) Relatively complete system, average integration (8–11) Loose system, low integration (4–7) Only partial solution (0–3)
Economic Benefits & Sustainability	15%	Return on investment analysis, energy saving, environmental friendliness	Outstanding economic benefits, green and sustainable (12–15) Some economic viability (8–11) Unclear economic viability (4–7) No relevant analysis (0–3)
Presentation & Team Collaboration	15%	Persuasiveness of presentation, demonstration of team division, collaboration in Q&A	Engaging presentation, cohesive team collaboration (12–15) Clear expression, good collaboration (8–11) Average expression, fair collaboration (4–7) Unclear expression, insufficient collaboration (0–3)

Track 4: AI+Materials Testing and Characterization

Dimension	Weight	Specific Content	Scoring Criteria
Detection Accuracy & Efficiency	30%	Algorithm accuracy, recall rate, detection speed, level of automation	High accuracy, fast speed, fully automated (25–30) Good accuracy, acceptable efficiency (18–24) Average accuracy, low efficiency (10–17) Poor accuracy, low efficiency (0–9)
Technical Completeness & Robustness	25%	System stability, data preprocessing capability, exception handling mechanism	Robust system, complete workflow, strong fault tolerance (20–25) Basically complete, some stability (13–19) Missing parts or vulnerabilities (6–12) Incomplete system (0–5)
Multi-Source Data Fusion Capability	20%	Ability to integrate and analyze multi-scale, multi-modal data comprehensively	Advanced fusion methods, in-depth analysis (16–20) Basic integration and analysis capability (11–15) Simple fusion, superficial analysis (6–10) No effective fusion achieved (0–5)
Industrial Application Adaptability	10%	Compatibility with production lines, deployment difficulty, meeting real-time requirements	Highly adaptable, easy deployment, meets real-time requirements (9–10) Basically adaptable, requires some adjustment (7–8) Moderate adaptability (4–6) Detached from industrial reality (0–3)
Presentation & Team Collaboration	15%	Persuasiveness of presentation, demonstration of team	Engaging presentation, cohesive team collaboration (12–15) Clear expression, good collaboration (8–11)

		division, collaboration in Q&A	Average expression, fair collaboration (4–7) Unclear expression, insufficient collaboration (0–3)
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Track 5: AI+Corporate Innovation and Applications

Dimension	Weight	Specific Content	Scoring Criteria
Needs Understanding & Alignment	30%	Accuracy in grasping industry/enterprise pain points, specificity of solution	Precisely grasps pain points, highly targeted solution (25–30) Good understanding of needs, relatively targeted solution (18–24) Average understanding, limited specificity (10–17) Misunderstood or disconnected from needs (0–9)
Innovation & Technology Integration	20%	Originality of solution, technology integration capability, level of cross-disciplinary fusion	Outstanding innovation, clever integration, good cross-disciplinary fusion (16–20) Some innovation and integration (11–15) Limited innovation, average integration (6–10) Lacks innovation (0–5)
Implementability & Execution Path	20%	Commercial/technical feasibility, reasonableness of resource requirements, clarity of phased planning	Clear implementation path, controllable resources, clear phases (16–20) Basically feasible, relatively clear planning (11–15) Some implementation obstacles (6–10) Difficult to implement (0–5)
Commercial Value & Market Potential	15%	Market size, profit model, competitive analysis, investment value	Significant value, clear model, clear potential (12–15) Some value, preliminary model formed (8–11) Unclear value, immature model (4–7) Lacks commercial consideration (0–3)
Presentation & Team Collaboration	15%	Persuasiveness of presentation, demonstration of team division, collaboration in Q&A	Engaging presentation, cohesive team collaboration (12–15) Clear expression, good collaboration (8–11) Average expression, fair collaboration (4–7) Unclear expression, insufficient collaboration (0–3)

Track 6: AI + Integrated Applications

Dimension	Weight	Specific Content	Scoring Criteria
Interdisciplinary Integration & Innovativeness	30%	Originality of project, novelty of concept, depth and cleverness of integration between AI and chosen field	Strong innovativeness, clever and deep integration (25–30) Clear innovation, reasonable integration (18–24) Some innovation, but conventional integration (10–17) Lacks innovation, forced technical application (0–9)
Technical Implementation & Completeness	25%	Reasonableness of technical approach, feasibility/efficiency of algorithm/model, system completeness, Demo effectiveness	Advanced technology, complete system, vivid demonstration, thorough validation (20–25) Feasible technology, basic functionality, clear demonstration (13–19) Deficiencies in technical approach, missing functions, simple demonstration (6–12) Technically infeasible, cannot be demonstrated or incomplete materials (0–5)
Application Value & Social Impact	20%	Project's ability to solve practical problems, potential user base/market size, positive	Significant application value, outstanding social benefit, high market potential (16–20) Some application prospects and social value (11–15)

		social/cultural/environmental impact	Limited scope, unclear value (6–10) Lacks clear application scenario or value (0–5)
Data & Ethical Compliance	10%	Legality of data sources, privacy protection measures, fairness and transparency of algorithm, consideration of potential social/ethical risks	Compliant data, privacy protection, clear ethical risk response (9–10) Clear data source, basic ethical considerations (7–8) Ambiguities regarding data or ethics (4–6) Obvious data or ethical issues (0–3)
Presentation & Team Collaboration	15%	Persuasiveness of presentation, demonstration of team division, collaboration in Q&A	Engaging presentation, cohesive team collaboration (12–15) Clear expression, good collaboration (8–11) Average expression, fair collaboration (4–7) Unclear expression, insufficient collaboration (0–3)

Notes:

1. Each judge will score independently. The final score will be the average of all judges' scores.
2. Any cheating behavior during the competition, such as plagiarism or misappropriation of others' work, will result in a score of zero. (Offenders will be subject to a formal reprimand, and prizes and certificates will be revoked. Subsequent rankings will be adjusted accordingly.)
3. In case of significant disagreement over a score, the Chair of the Judging Committee will convene the judges for a collective decision.

IX. Contact Us

Tel: +86 10-6277 1624

Email: jiali-zhao@mail.tsinghua.edu.cn

X. Supplementary Notes

1. Supporters

China Baowu Steel Group Co., Ltd.

Contemporary Amperex Technology (Hong Kong) Limited

Beijing NAURA Microelectronics Equipment Co., Ltd.

Ansteel Group Co., Ltd.

China Telecom Corporation Limited, Beijing Branch

Chuangcai Shenzao (Tianjin) Technology Co., Ltd.

(More to be added...)

2. Innovation Bootcamp Supporters

Shuimu Tsinghua Alumni Seed Fund

Shuimu Incubator

3. Computing Power Supporters

Beijing Volcano Engine Technology Co., Ltd.

Alibaba Cloud Computing Co., Ltd.

Beijing Paratera Tech Corp., Ltd.

Sugon Intelligent Computing Information Technology Co., Ltd.

4. Venture Capital & Media Supporters

People.cn Venture Capital Co., Ltd.

5. The final interpretation rights of this competition belong to the Competition Organizing Committee.